EXPERIMENT 07

Lake_Wave_Numerical_Analysis_Report

THAKOR PAL(23BME002)

1. Aim

To simulate the propagation of a small-amplitude surface wave in a lake using the linearized one-dimensional shallow-water equations, and to compare the numerical solutions obtained using the Upwind and Lax–Wendroff schemes with the analytical solution.

2. Governing Equations

The supplied solver models linearized, one-dimensional shallow-water motion over a constant mean water depth. The governing equations are:

$$\partial\eta/\partial t + H\,\partial u/\partial x = 0$$

$$\partial u/\partial t + g\,\partial\eta/\partial x = 0$$

Here, η is the free-surface displacement, u is the depth-averaged horizontal velocity, H is the mean water depth, and g is gravitational acceleration. The wave speed is $c = \sqrt{(gH)}$. For the supplied parameters, $c = \sqrt{(9.81 \times 1.0)} \approx 3.132$ m/s.

3. Initial Condition and Analytical Reference

The initial free-surface disturbance is a Gaussian pulse centred at $x = 10$ m. Its maximum initial amplitude is 0.02 m and its width parameter is $\sigma = 0.5$ m:

$$\eta(x, 0) = A \exp[-(x - x_0)^2 / (2\sigma^2)]$$

The analytical reference used by the solver represents the initial pulse as two equal-amplitude Gaussian waves travelling in opposite directions:

$$\eta(x,t) = (A/2) \{ \exp[-(x - x_0 - ct)^2/(2\sigma^2)] + \exp[-(x - x_0 + ct)^2/(2\sigma^2)] \}$$

4. Numerical Methods

The solver advances the two characteristic variables of the linearized shallow-water system in time. Two finite-difference methods are compared:

4.1 Upwind scheme

The Upwind method uses a one-sided spatial difference selected according to the direction of wave propagation. It is robust for advective transport, but its numerical diffusion can smooth and reduce the peak of a moving pulse.

4.2 Lax–Wendroff scheme

The Lax–Wendroff method uses a centred second-order update with a second-difference correction. It generally provides better phase and amplitude accuracy for smooth wave propagation, although dispersive effects may occur near sharp gradients.

5. Simulation Parameters

The notebook cell displays RUN_MODE = “quick”. The solver receives the QUICK_ARGS list because the run-mode selection chooses QUICK_ARGS when RUN_MODE is quick. The parameter names in the notebook wrapper are passed positionally to the solver; the solver interprets the second value as domain length, the fifth as Gaussian width, and the sixth as the initial pulse centre.

Parameter	**Value used by solver**	**Description**
Grid points, nx	81	Number of points in x direction
Domain length, L	20.0 m	Solver's second positional argument
Mean depth, H	1.0 m	Still-water depth
Initial amplitude, A	0.02 m	Peak of initial Gaussian disturbance
Gaussian width, σ	0.5 m	Width parameter of initial pulse
Initial centre, x_0	10.0 m	Centre of the initial pulse
Final simulation time	0.1 s	Quick run duration
CFL number	0.8	Courant number
Gravity, g	9.81 m/s²	Gravitational acceleration
Output interval	10 steps	VTK/profile output interval
Vertical visualization scale	30	Scale used for VTK visualization
VTK vertical points	15	Number of points in visualization direction

6. Results and Discussion

6.1 Surface elevation field

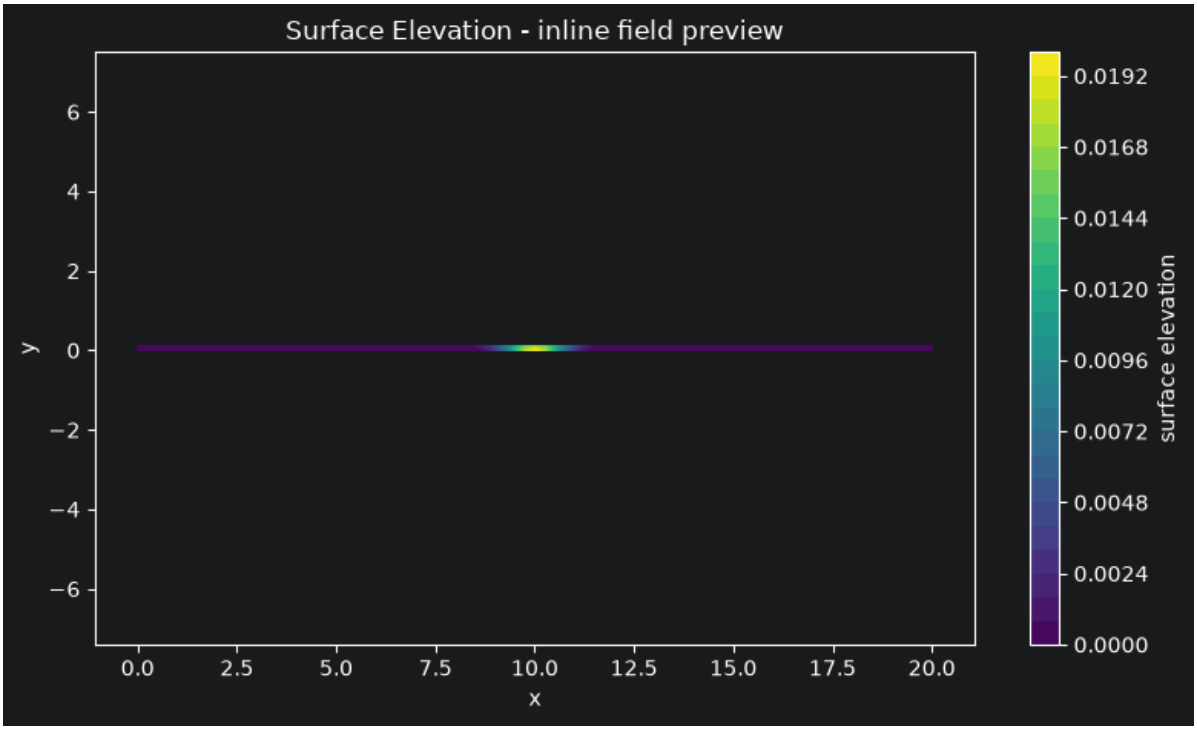


Figure 1. Inline preview of the surface-elevation field.

The field preview shows the surface disturbance concentrated around the centre of the domain, near x = 10 m. Away from the pulse, the surface elevation is close to zero. The plotted field is displayed as a very thin horizontal band because the free-surface displacement is small compared with the visualization extent. The colour scale indicates that the largest elevation occurs in the central disturbed region.

6.2 Final wave-profile comparison

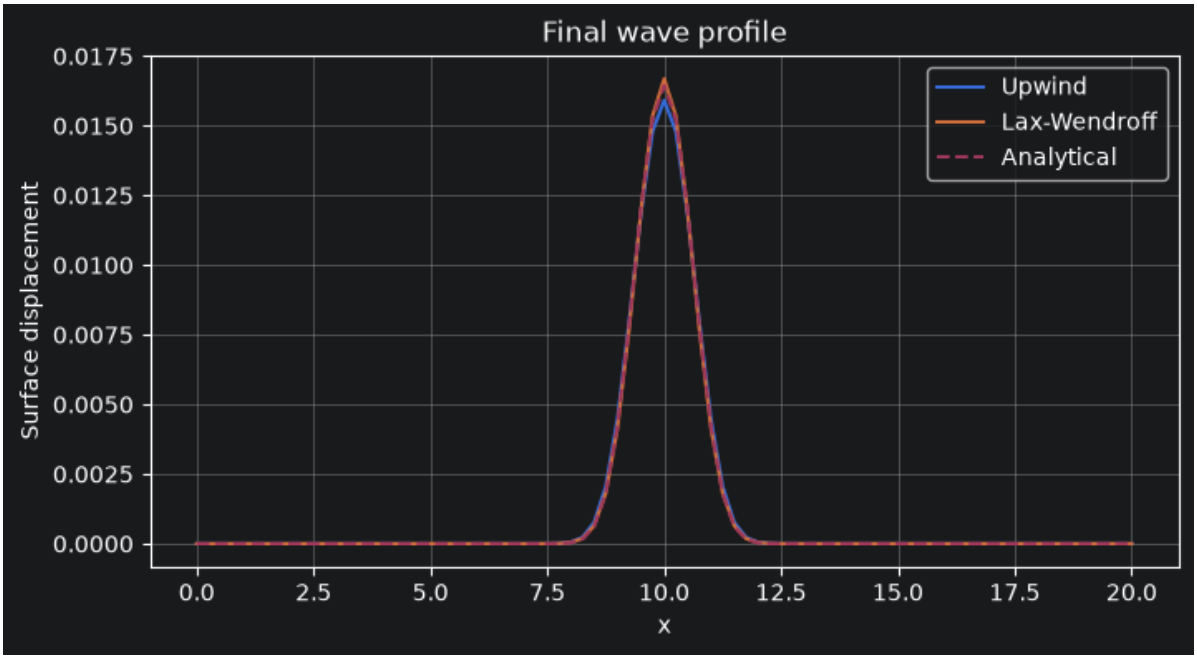


Figure 2. Final surface-displacement profiles from Upwind, Lax–Wendroff, and the analytical reference.

All three profiles retain a narrow, approximately symmetric pulse centred near x = 10 m. The numerical curves follow the analytical reference closely. The Upwind curve has a slightly lower and smoother peak, which is consistent with numerical diffusion. The Lax–Wendroff curve lies closer to the analytical peak over most of the pulse and gives the closer visual match in this result.

6.3 Error evolution

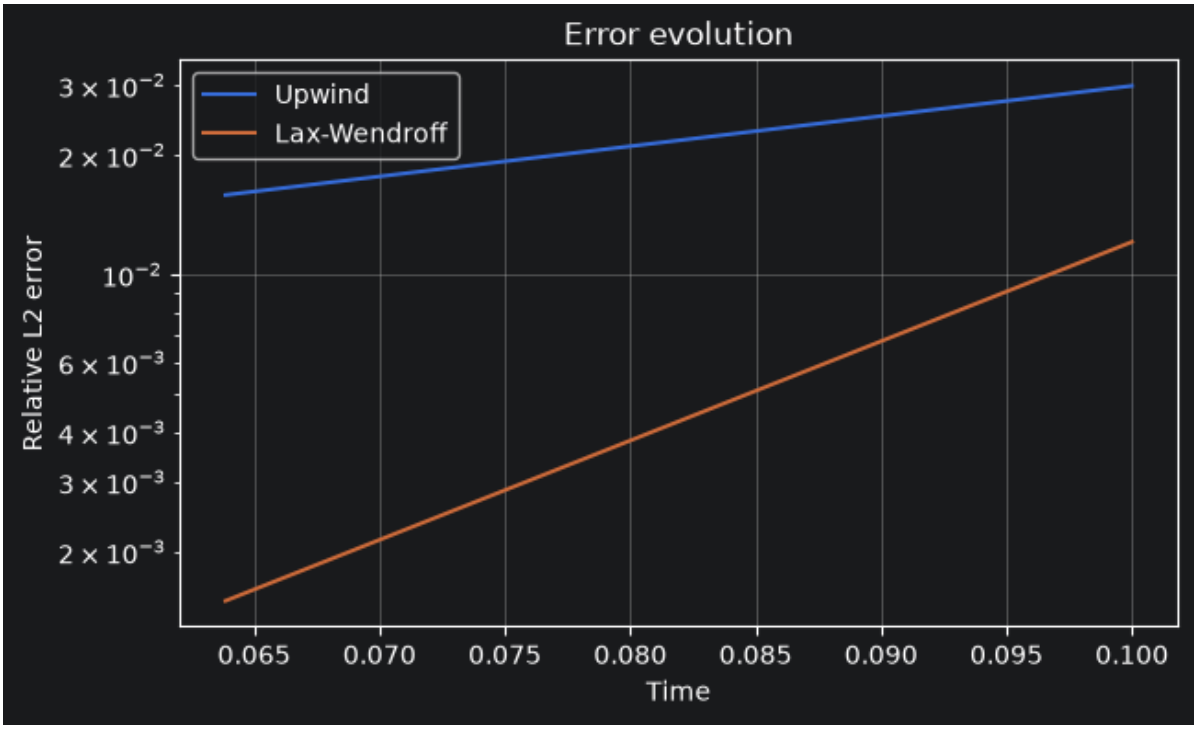


Figure 3. Relative L2 error versus time for the two numerical schemes.

The relative L2 error increases with time for both methods over the plotted interval. However, the Upwind error remains higher than the Lax–Wendroff error throughout the displayed range. At the final plotted time of approximately 0.1 s, the graph indicates an Upwind error of about 3×10^{-2} and a Lax–Wendroff error of about 1.2×10^{-2}. These are approximate readings from the plotted graph, not exact values from the solver summary file. The error trend indicates that Lax–Wendroff is more accurate for this smooth pulse and selected grid/time-step settings.

7. Quantitative Comparison

Measure	**Upwind**	**Lax–Wendroff**
Final relative L2 error (from plot)	≈ 0.030	≈ 0.012
Relative error trend	Higher throughout	Lower throughout
Peak profile	Slightly reduced	Closer to analytical peak
Observed numerical behaviour	More smoothing	Better profile agreement

8. Physical Interpretation

The initial disturbance behaves as a small-amplitude gravity wave. In the linearized model, the wave speed depends on the mean depth and gravity, rather than on the wave amplitude. The analytical solution splits the initial Gaussian disturbance into two components travelling in opposite directions. During the short quick-run interval, the pulse remains close to the centre and retains its overall shape. The close agreement between the numerical profiles and the analytical reference indicates that both schemes capture the main wave behaviour. The difference in error is numerical: the Upwind method introduces more smoothing, while Lax–Wendroff preserves the pulse shape more accurately for this case.

9. Limitations of the Model

- The model uses linearized one-dimensional shallow-water equations; nonlinear wave effects are not represented.
- The mean water depth is constant, so varying bathymetry and complex lake geometry are not included.
- The simulation is one-dimensional and cannot represent lateral spreading or two-dimensional circulation.
- The model does not include viscosity, turbulence, wind forcing, bottom friction, or other dissipative effects.
- The displayed quick run ends at 0.1 s; longer-time behaviour is not established by these plots.
- The field preview uses a visualization scaling factor, so its displayed vertical geometry should not be interpreted as an unscaled physical water-body shape.
- Exact final error and mass-conservation values should be taken from lake_wave_summary.txt or lake_wave_history.csv; those numerical text outputs were not included with the supplied screenshots.

10. Conclusion

The one-dimensional linearized shallow-water wave was successfully simulated using Upwind and Lax–Wendroff finite-difference schemes. Both numerical profiles agree closely with the analytical Gaussian-wave reference. The error plot shows that Lax–Wendroff produces a lower relative L2 error than Upwind over the simulated time interval. The Upwind solution exhibits greater numerical smoothing, while Lax–Wendroff preserves the wave profile more accurately for the selected quick-run parameters. Therefore, Lax–Wendroff is the more accurate of the two methods for the present smooth-wave test.